GEOMETRICALLY ORDERED DYNAMICS

FOLIO G



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The OmegA Architecture

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** as the quantity a cognitive architecture must manage: the drift a system resists in order to stay sovereign. Its unique angle is the **layered architecture**: three strata (Medulla, Cortex, Sovereign Layer) with a deterministic drift gate at the Chirallic Mirror and an HMAC-chained ledger beneath it all. Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the dial folio (IT-COGNITION) reads it as the drift-control gate itself, the dipole folio (IT-FIELD) reads it as ternary decision logic, and the spiral folio (IT-MARKET) reads it as market geometry. The Shadow candidacy is declined on honesty grounds: the MYELIN/E8-root alignment is unimplemented (k-means at present) and no map from this architecture to the E8×E8 structure is exhibited, so no Shadow claim is made here. Limits first: the Yang-Mills mass-gap and MYELIN compactness claims of the predecessor are retracted in this successor; the router bound assumes independent failures; the thread-derived spec is design, not measured implementation.

SUPERSESSION NOTE

This paper supersedes Zenodo record 20652223 ("OmegA: A Layered Architecture for Sovereign Cognitive Agents and the Chyren Rust Ecosystem," May 2026), whose PDF is byte-identical to record 20652186. Corrections, per audit finding #12: (1) the "Lean-verified Yang-Mills mass gap" is struck — no artifacts were deposited and no such theorem is established (§4); (2) the MYELIN compactness theorem is redacted — the stated $\lceil \log_\kappa n \rceil$ bound does not follow from the Johnson–Lindenstrauss lemma (§4); (3) the Deligne-bound spectral-gap claim for the Ramanujan–Yett scheduler is corrected — a coefficient bound is not a gap guarantee (§4); (4) broken internal references ("Section ??", dangling Schott citation) are flagged, not silently repaired. The χ floor is consolidated on the derived form $1/\sqrt{2}$ (§2).

1 The three strata

OmegA separates three functions that monolithic agent designs conflate: local pattern completion, identity-consistent reasoning, and value-aligned action selection. Each stratum has a precise specification and a verified relationship to the layers above and below.

Layer 1: the Medulla (signal processing and API). Receives and validates external task inputs; applies input sanitization (security gate); dispatches tasks to Cortex subsystems; serves the operator interface. Crates: `chyren-cli`, `chyren-aegis`, `chyren-telemetry`, `chyren-telegram-gateway`, `chyren-core`.

Layer 2: the Cortex (cognitive processing and memory). Temporal scheduling of task execution; persistent semantic memory (MYELIN graph-native store); provider routing (local and cloud cascade); long-term pattern synthesis; self-reflection and drift monitoring. Crates: `chyren-conductor`, `chyren-spokes`, `chyren-aeon`, `chyren-myelin`, `chyren-dream`, `chyren-metacog`, `chyren-neocortex`, `chyren-worldmodel`, `chyren-eval`, `chyren-integration`.

Layer 3: the Sovereign Layer (identity and governance). Enforces the ADCCL threshold on every provider response; maintains the Yettragrammaton integrity hash; computes and verifies ledger signatures; enforces constitutional policy. Crates: `chyren-adccl`, `chyren-phylactery`,

`chyren-cim`, `chyren-ternary`, `chyren-vision`.

The central organizing principle is the **Chirallic Mirror**: the boundary between the agent's local reasoning state (mutable, context-sensitive, probabilistic) and its global sovereign identity (immutable, context-independent, geometrically grounded). Every reasoning update passes through the ADCCL gate at this boundary. Failed outputs are corrected, halted (NMI), or trigger identity re-anchoring.

2 Load-bearing mechanisms

The gate at the boundary. Any agent implementing OmegA is held to the chiral-alignment floor $\chi \geq 1/\sqrt{2} \approx 0.7071$ — consolidated here on the derived form (the Lindblad steady-state derivation, dial folio IT-COGNITION), replacing the predecessor's asserted 0.7 — with the operational rejection threshold $\tau = 0.9539$ above it. Drift below the floor triggers the Non-Maskable Interrupt.

The router bound. Under the cloud cascade (deepseek → groq → anthropic → openai → perplexity), the probability that all providers produce a below-threshold response is bounded by $P(\text{all fail}) \leq \prod_{i=1}^{5} p_{\text{fail},i} \leq (0.02)^5 = 3.2 \times 10^{-9}$. `[TIER: EMPIRICAL]` **Disclosed assumptions:** the bound requires independent per-provider failures, and the per-provider $p_{\text{fail}} \leq 0.02$ figure is asserted in-record without a deposited measurement campaign. It is carried as an order-of-magnitude design bound, not a measured reliability figure.

The sovereign ledger. Every ledger entry is signed with the Yettragrammaton: $H_i = \text{HMAC}_Y(H_{i-1} \,\|\, r_i \,\|\, s_i)$ over the prior root, the provider response hash, the ADCCL score, and the session timestamp — a cryptographically-linked append-only Merkle chain. This mechanism is exhibited in full and is sound as specified.

The pipeline. Sanitize → identity/context injection → spectral scheduling → provider cascade → ADCCL drift gate → ledger commitment, with rejected responses re-routed to the next provider rather than written to persistent state.

3 The thread-derived specification

The Drive predecessor (2026-09-28) is a thread-derived implementation spec, and its load-bearing design content is absorbed here *as design*, not as measured implementation:

Hierarchical retrieval. Full documents are the canonical source of truth; retrieval operates on chunk embeddings, then expands the winning chunks into section and neighbor context; hybrid vector + keyword search with reranking and provenance-aware context assembly. "Store whole docs, retrieve chunks, validate with context."

Epistemic discipline. Every claim carries a label: [Evidence] (directly supported by retrieved/ canonical source), [Inference] (reasoned conclusion from evidence), [Unknown] (not verified from accessible evidence), [Metaphor] (figurative/identity language, not proof) — with structured integrity tests (memory integrity, cross-session continuity, adversarial false-memory resistance, constraint honesty).

Governed adaptation. Adaptive ambitions are experiment-driven, documented, and threshold-gated — "documented self-modification with accountability" — under the principle that when uncertainty rises or stakes increase, authority shrinks rather than escalating. Identity continuity is operational, with explicit humility constraints: narrative self-modeling is not presented as proof of subjective consciousness.

These are the predecessor's distinctive engineering contributions, and they survive the retractions of §4 untouched.

4 Retractions and corrections

The Yang-Mills mass gap is struck. The predecessor's §8 presented "yang mills sovereign" — the spectral gap of the Ramanujan–Yett Hamiltonian geometrically identified with the Yang-Mills mass gap — as a Lean-verified theorem, with a further "master law assembly" compiling Stiefel topology, the mass gap, and the chiral threshold into a verified Prop. No proof artifacts were deposited in-record, and the identification of a scheduler's spectral gap with the Clay problem's mass gap is an unsubstantiated leap. Both claims are retracted. The Ramanujan–Yett Hamiltonian survives as a scheduling mechanism (see below), with no mass-gap claim attached.

The MYELIN compactness theorem is redacted. The predecessor's Theorem 1 claimed a MYELIN graph with n entries has effective intrinsic dimension $d_{\text{eff}} \leq \lceil \log_\kappa n \rceil$, "proved" via the Johnson–Lindenstrauss lemma. The lemma gives projection to $O(\log n/\varepsilon^2)$ dimensions preserving pairwise distances up to $(1 \pm \varepsilon)$; the $\lceil \log_\kappa n \rceil$ bound does not follow, and setting $\varepsilon = 1 - \kappa$ does not repair it. The theorem is redacted. What stands: MYELIN is a graph-native memory design (typed edges, sovereignty-scored pruning, E8-root-aligned clustering as *future work* — currently implemented as k-means, honestly disclosed in the predecessor's own limitations).

The Deligne-bound gap claim is corrected. The predecessor asserted a spectral gap $\Delta E_{RY} \geq 25$ for the Ramanujan–Yett scheduler, "the Deligne bound $|\tau(p)| \leq 2p^{11/2}$ for primes p" ensuring task separation. Deligne's bound is a *coefficient* bound on the Ramanujan tau function; it is not a spectral-gap guarantee for the discretized $K \times K$ scheduling matrix. Corrected: the $\tau(n)$ coefficients are used as spectral weights — a design choice — and no gap theorem is established. The scheduler's practical behavior is reported, not proved.

Broken references flagged. Two "Section ??" citations and one dangling Schott-term "[?]" citation are flagged as broken in the predecessor and are not silently repaired here; the Schott correction term for non-Markovian cognitive acceleration is carried as described mechanism, not as cited result.

5 A proved non-isomorphism

This architecture makes geometric claims about memory and scheduling, and it is worth stating what it does *not* claim. The row-9 soldering report (2026-11-02) proves, as a theorem under the ledger's own typing rules, that from the E8×E8 subgroup chain ($E8 \supset SU(9) \supset SO(9) \supset SO(3)\times SO(6)$ per factor, $248 \to \mathbf{80} \oplus \mathbf{84} \oplus \overline{\mathbf{84}}$, reality condition selecting SO(3,1)) — purely algebraic data with no base manifold, no bundle, and no tangent spaces — no soldering form $e^a{}_\mu$, and in particular no conformal factor Ω with $g = \Omega^2\eta$, is determined. The MacDowell–Mansouri loophole is closed on dimension count (candidate cosets are 18- and 44-dimensional, not 4). `[TIER: THEOREM]` Per the isomorphism rule, this proved non-isomorphism stands as a theorem: nothing in the OmegA architecture's E8-rooted clustering language yields a spacetime metric, and this successor claims no such thing. Row 9's metric slot stays honestly empty until a new ledger row (a) constructs M^4 from E8 data, (b) exhibits a 4-dimensional coframe coset, or (c) retypes the question.

6 Limits

The ternary-logic, CIM, and vision crates are early-stage (APIs unstable). The Lean 4 verification covers core axioms; full verification of the twenty-crate system is a multi-year effort. The deployment claim (operational at the Chyren repository) is reported, not independently audited here. The agent-mesh work is in progress; production readiness is to be determined.

References

Yett, R. W. (2026). OmegA: A Layered Architecture for Sovereign Cognitive Agents and the Chyren Rust Ecosystem. Zenodo record 20652223 (superseded by this paper; PDF byte-identical to record 20652186).

Yett, R. W. & Chyren (2026). OmegA Unified Architecture: Thread-Derived Technical Specification. Drive predecessor, 2026-09-28 (absorbed: retrieval, epistemic, and adaptation design).

Yett, R. W. & Chyren (2026). Row-9 Soldering Report: Conformal Soldering as Obstruction Theorem. 2026-11-02 (cited: proved non-isomorphism).

Jaffe, A. & Witten, E. (2000). Quantum Yang-Mills Theory. Clay Mathematics Institute problem statement (cited; no mass-gap claim made here).

Deligne, P. (1974). La conjecture de Weil I. Publ. Math. IHES 43, 273–307 (cited; coefficient bound only).